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Augmented Reality for Corporate Sustainability Reporting: An Interactive Carbon Footprint System

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***Abstract*—Corporate sustainability reports are often difficult for non-expert audiences to understand because they are typically presented as static, number-dense PDF documents. This paper presents an interactive Augmented Reality (AR) system that transforms Herc Rentals' Corporate Citizenship Report data into spatially anchored 3D visualizations on a commodity Android smartphone. Developed using Unity and AR Foundation, the system detects the corporate logo and dynamically renders three scaled emission spheres representing Scope 1, Scope 2, and Scope 3 greenhouse gas emissions, totaling 358,841 MT CO_2e in 2024. By tapping a sphere, users can access contextual panels that provide category-level breakdowns and carbon offset equivalencies derived from U.S. Environmental Protection Agency (EPA) greenhouse gas equivalency factors. A preliminary user evaluation with six participants ($N = 6$), using the System Usability Scale (SUS) and NASA Task Load Index (NASA-TLX), produced a mean SUS score of 77.9, surpassing the target threshold of 75, along with generally low workload ratings across NASA-TLX dimensions. These findings suggest that AR-based sustainability visualization can improve comprehension, engagement, and environmental awareness in corporate reporting contexts.**

***Index Terms*—augmented reality, sustainability reporting, carbon footprint visualization, AR Foundation, human-computer interaction, greenhouse gas emissions**

I. Introduction

Corporate sustainability reporting has become an important mechanism for environmental accountability and stakeholder communication. Many organizations now publish Corporate Citizenship Reports (CCRs) or sustainability reports that summarize greenhouse gas (GHG) emissions, energy consumption, waste generation, and water usage. Although these reports contain valuable information, they are typically presented as static, number-dense PDF documents that are difficult for non-expert audiences to interpret. For example, a value such as 161,635 MT CO_2e for Scope 1 emissions may be precise, yet it remains abstract and cognitively inaccessible to readers.

This communication gap has practical consequences. Employees, customers, investors, and regulators who cannot readily interpret Scope 1–3 boundaries, category-level breakdowns, or offset equivalencies may be less likely to engage meaningfully with corporate sustainability disclosures. Prior work in environmental psychology suggests that people respond more strongly to embodied and spatially grounded representations than to abstract numerical information and that immersive experiences can influence both environmental locus of control and pro-environmental behavior [1]. These findings motivate the use of Augmented Reality (AR) as a medium for sustainability communication.

AR enhances the physical world by overlaying digital content onto real-world surroundings in real time. Recent mobile AR frameworks, including Unity's AR Foundation, support robust image tracking and plane detection on commodity smartphones, making it possible to situate digital environmental information directly around physical artifacts such as a company's printed logo [2]. Prior studies have explored AR in several sustainability-related contexts, including promoting environmental action through game-based interaction [3], representing numerical information through immersive visual metaphors [4], visualizing building energy consumption and indoor environmental quality [5], and encouraging pro-environmental behavior through interactive AR experiences [6]. Broader reviews also suggest that virtual and augmented reality technologies have strong potential to support environmental sustainability and education by making complex concepts more accessible and engaging [7], [8].

Despite this progress, there remains limited work on AR systems designed specifically for corporate-level GHG disclosure, particularly systems that visualize Scope 1, Scope 2, and Scope 3 emissions using real organizational data. Existing AR sustainability applications have focused primarily on education, gaming, or built-environment scenarios [3], [5], [7], [8], rather than on interactive interpretation of corporate emissions reports. To address this gap, we present a logo-anchored AR system that transforms Herc Rentals' corporate carbon data into spatially situated 3D visualizations on a commodity Android smartphone. In addition to displaying emissions by scope, the system incorporates U.S. Environmental Protection Agency (EPA) greenhouse gas equivalency factors to translate large carbon values into more intuitive comparisons, such as tree planting, electric vehicle charging, and diesel fuel usage [9]. Accordingly, this work investigates whether spatially anchored AR visualization can improve the interpretability of Scope 1–3 emission magnitudes for non-expert stakeholders, whether a logo-anchored AR carbon visualization interface can be used effectively without prior training, and whether EPA-based offset equivalencies can further enhance the in-

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terpretability of large corporate emissions values. The main contributions of this paper are as follows:

- We present a novel AR-based representation of corporate carbon footprint data using a three-bubble, scope-oriented visualization anchored to a physical organizational logo and implemented with Unity and AR Foundation.
- We integrate multi-year Herc Rentals CCR emissions data (2022–2024) with EPA greenhouse gas equivalencies, enabling large emission values, including Scope 3 emissions up to 183,425 MT CO_2e, to be interpreted through relatable comparisons such as tree planting, EV charging, and diesel fuel consumption.
- We report a formative usability and workload evaluation with six participants ($N = 6$) using the System Usability Scale (SUS) and NASA Task Load Index (NASA-TLX), obtaining a mean SUS score of 77.9 and generally low cognitive and physical workload ratings.

II. Related Work

A growing body of work has examined the use of AR and XR for sustainability, environmental education, and pro-environmental decision support. Ladykova *et al.* reported that AR-based environmental education often improves learner motivation, engagement, and conceptual understanding, though evidence for long-term behavioral change remains limited [7]. Cosio *et al.* similarly found that immersive technologies can support environmental awareness and action, while noting that many systems remain narrow in scope and lack formal usability evaluation [8]. At the application level, Wang *et al.* showed that an AR game could positively influence environmental attitudes and behavioral intentions [3], and Kim and Lee demonstrated that AR interaction can encourage pro-environmental decision-making in everyday contexts [6]. Collectively, these studies suggest that AR is both engaging and potentially effective for sustainability communication.

Recent work has also highlighted the value of AR for presenting complex environmental and numerical information in more accessible forms. Marques *et al.* introduced immersive unit visualization to communicate abstract quantities in situ, showing that interactive 3D representations can improve scale comprehension [4]. Sun *et al.* applied AR to waste-sorting education through a gamified recycling assistant, reporting improvements in user awareness and task accuracy [10]. In the built-environment domain, An *et al.* linked live building energy and indoor environmental quality data to AR overlays, observing improved user understanding and reductions in energy consumption [5]. These findings indicate that AR is particularly effective when abstract sustainability metrics are translated into situated, interactive visual forms.

A parallel line of research has explored mobile sensing and carbon-related environmental assessment. Magnuson *et al.* reviewed mobile-device-based approaches for forest mensuration, highlighting the growing practicality of consumer-grade platforms [11]. Tatsumi *et al.* demonstrated that the iOS LiDAR-based *ForestScanner* can achieve high agreement with traditional tree inventory methods [12], and Sae-ngow *et al.* validated smartphone LiDAR for urban forest carbon assessment [13]. Complementary work has used machine learning and aerial sensing for large-scale carbon estimation, including DeepAR-based sequestration modeling [14] and drone-based carbon storage assessment [15]. Although not focused on corporate reporting, these studies reflect a broader trend toward using advanced visualization and sensing technologies to make carbon-related information more measurable, interpretable, and actionable.

Recent work has strengthened the case for interactive and immersive approaches to sustainability communication. Zhang *et al.* introduced *Living Sustainability*, an in-context system that helps non-experts interpret environmental impact data through contextualized visualizations rather than expert-oriented workflows [16]. Rega *et al.* further showed that AR can support sustainability practices in applied industrial settings, extending its use beyond education-focused scenarios [17]. A recent scoping review by Simon *et al.* similarly found that AR generally improves engagement and learning in environmental education, though gaps remain in evaluation rigor and participant diversity [18]. Related immersive research has demonstrated comparable benefits: Huber *et al.* used VR to present complex CO_2 data in intuitive spatial and temporal forms [19], and Alex *et al.* reported that AR and VR can both enhance engagement with climate-action content [20].

Despite these advances, most AR sustainability systems have focused on education, behavioral interventions, or ecological measurement. To our knowledge, no prior work has applied logo-anchored AR visualization to corporate Scope 1, Scope 2, and Scope 3 greenhouse-gas disclosures while integrating offset equivalencies and evaluating usability and workload. The present work addresses this gap by applying AR to real corporate reporting data.

III. Methodology

A. System Overview

The AR Carbon Footprint Visualizer is designed to address a key usability challenge in corporate sustainability reporting: transforming a static printed corporate logo into an interactive portal for multi-scope greenhouse gas (GHG) information. When the Herc Rentals logo is detected through AR Foundation image tracking, the application instantiates three floating emission spheres corresponding to Scope 1, Scope 2, and Scope 3 emissions, with sphere sizes scaled proportionally to annual emission magnitudes. Selecting any sphere reveals a contextual information panel that presents category-level emission breakdowns along with intuitive offset equivalencies. In addition, a year slider allows users to compare data across the 2022–2024 reporting period, while an offset toggle enables switching among tree-planting, EV-charging, and diesel-fuel equivalency views.

As illustrated in Fig. 1, the system architecture consists of four main layers. The *Data Layer* stores CCR emission values and precomputed U.S. Environmental Protection Agency (EPA) offset equivalencies in a structured JSON format. The

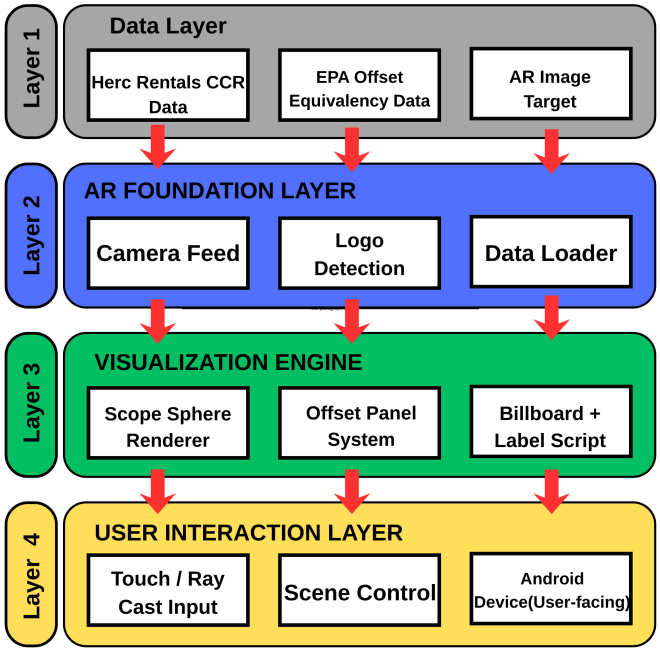


Fig. 1: System architecture of the proposed AR Carbon Footprint Visualizer, showing the data, AR tracking, visualization, and user interaction layers.

AR Foundation Layer is responsible for image-target recognition and world-space anchoring of the detected corporate logo. The *Visualization Engine* manages sphere instantiation, proportional scaling, billboard-style information panels, and the year-based timeline logic. Finally, the *User Interaction Layer* supports touch-based ray casting, feature toggling, and interaction event logging. The system is implemented in Unity using AR Foundation and is designed to run on Android smartphones.

B. Data Preparation and Emission Dataset

Scope 1, Scope 2, and Scope 3 emissions data were compiled from Herc Rentals' publicly available Corporate Citizenship Reports (CCRs) for the years 2022, 2023, and 2024. The scope definitions follow the greenhouse gas accounting boundaries described in the IPCC Guidelines for National Greenhouse Gas Inventories [21]. For each reporting year, total emissions and available category-level breakdowns were manually extracted from the source documents, converted into a machine-readable JSON structure, and verified through cross-checking against the original CCR tables. The resulting multi-year dataset is summarized in Table I.

Some Scope 3 categories were not consistently reported across all years. In particular, categories 3 and 7 were unavailable in the 2022 report and were therefore recorded as not reported (N/R). To support real-time interactive visualization without incurring repeated on-device computations, offset equivalencies were precomputed for each scope and year using U.S. Environmental Protection Agency (EPA) greenhouse gas equivalency factors [9]. These values were stored as static lookup entries in the JSON dataset and retrieved at runtime by the visualization engine.

TABLE I: Herc Rentals GHG Emissions by Scope and Year (MT CO_2e)

Category	Scope	2024	2023	2022	Δ 22→24
Total Scope 1	Scope 1	161,635	148,668	133,955	+20.7%
Diesel	—	100,391	92,978	82,529	+21.6%
Gasoline	—	46,627	42,256	37,769	+23.4%
Natural Gas	—	14,329	13,434	13,657	+4.9%
Total Scope 2	Scope 2	13,781	13,796	13,427	+2.6%
Total Scope 3	Scope 3	183,425	166,666	80,526	+127.8%*
Total Scope 1+2	—	175,416	162,464	147,382	+19.0%

*Scope 3 increase is partly attributable to expanded category reporting in 2023–2024.

C. Bubble Scaling Model

To visually encode emission magnitude, each scope is represented as a sphere whose rendered radius is proportional to its annual GHG total. Let $E_i \in \mathbb{R}_{\geq 0}$ denote the emission magnitude of scope $i \in \{1, 2, 3\}$, measured in MT CO_2e. Let $s_{\min}$ and $s_{\max}$ represent the minimum and maximum allowable sphere radii in Unity world-space units. The rendered radius s_i is defined as

$$s_i = s_{\min} + (s_{\max} - s_{\min}) \frac{E_i}{\max(E_1, E_2, E_3)}. \quad (1)$$

This normalization ensures that the largest scope for a given year is always rendered at full scale $s_{\max}$, while the remaining scopes are scaled proportionally. The formulation preserves relative magnitude relationships across scopes while maintaining a lower visual bound for interaction, thereby preventing smaller values from becoming too small for reliable touch selection.

For the 2024 dataset, Scope 3 constitutes the maximum emission value with 183,425 MT CO_2e and is therefore rendered at $s_{\max}$. Under the same normalization, Scope 1 is displayed at approximately 88% of the maximum scale, whereas Scope 2 remains visibly smaller but above the minimum interaction threshold.

D. Offset Equivalency Computation

Because raw emissions values at the scale of tens or hundreds of thousands of metric tons can be difficult for non-expert users to interpret, the system augments each scope with intuitive offset equivalencies. For each scope i, three equivalency quantities are computed: the number of trees required to sequester the same amount of carbon over one year, the equivalent number of electric vehicle (EV) charging events, and the equivalent gallons of diesel fuel.

Let E_i denote emissions in MT CO_2e. Since the EPA factors are expressed in kilograms of CO_2e, emissions are first converted to kilograms via $1000E_i$. The equivalency mappings are then computed as

$$T_i = \frac{1000E_i}{r_{\text{tree}}}, \quad (2)$$

$$V_i = \frac{1000E_i}{r_{\text{EV}}}, \quad (3)$$

$$D_i = \frac{1000 E_i}{r_{\text{diesel}}}, \tag{4}$$

where T_i is the annual tree sequestration equivalent, V_i is the EV charging equivalent, and D_i is the diesel fuel equivalent. The constants are defined as $r_{\text{tree}} = 60.4775$ kg CO_2e sequestered per tree per year, $r_{\text{EV}} = 373.35$ kg CO_2e per MWh of EV-equivalent electricity, and $r_{\text{diesel}} = 10.18$ kg CO_2e per gallon of diesel, based on U.S. EPA greenhouse gas equivalency factors and EPA eGRID data [9].

To minimize runtime overhead, all equivalency values were precomputed offline and stored in lookup tables indexed by year and scope. Table II presents the precomputed equivalencies for the 2024 dataset.

TABLE II: Precomputed 2024 Offset Equivalencies by Scope

Scope	Emissions (MT CO_2e)	Tree Equiv.	EV Charge Equiv.	Diesel (gal)
Scope 1 (Direct)	161,635	2,672,649	432,932	15,877,701
Scope 2 (Electricity)	13,781	227,870	36,912	1,353,733
Scope 3 (Value Chain)	183,425	3,032,948	491,295	18,018,173
Total	358,841	5,933,467	961,139	35,249,607

E. Touch Interaction and Ray Casting

Interactive selection is implemented through a touch-driven ray-casting mechanism. At each frame, when a valid pointer event is detected, a ray is projected from the camera center through the screen-space interaction coordinate. The resulting world-space ray is parameterized as

$$\mathbf{r}(t) = \mathbf{o} + t\hat{\mathbf{d}}, \quad t \geq 0, \tag{5}$$

where $\mathbf{o}$ is the camera origin and $\hat{\mathbf{d}}$ is the normalized viewing direction derived from the projection geometry and tap location. Intersection tests are then performed against the colliders of tracked scene objects.

If the ray intersects the logo collider, the system toggles instantiation of the three scope spheres. If the ray intersects one of the scope spheres, the corresponding information panel is toggled. This interaction model is idempotent: selecting a sphere when its panel is absent creates the panel, while selecting it again removes the panel. Such a design simplifies the interaction grammar and reduces ambiguity during exploratory use.

To support both desktop debugging and mobile deployment, the implementation accommodates mouse-based input within the Unity Editor and multi-touch input on Android devices via conditional input handling.

F. Panel Orientation and Billboarding

To maintain readability under arbitrary user motion, all information panels employ a billboarding strategy that continuously aligns them with the viewing camera. Let $\mathbf{p}_i$ denote the world-space position of panel i, and let $\mathbf{p}_{\text{cam}}$ denote the camera position. The forward-facing unit vector of the panel is updated as

$$\hat{\mathbf{f}}_i = \frac{\mathbf{p}_{\text{cam}} - \mathbf{p}_i}{\|\mathbf{p}_{\text{cam}} - \mathbf{p}_i\|}. \tag{6}$$

The panel rotation is then set such that its local forward axis is aligned with $\hat{\mathbf{f}}_i$. This strategy ensures that textual information remains front-facing and legible, even as the user changes viewpoint or physically walks around the AR scene.

G. Usability Evaluation Protocol

A formative usability study was conducted to assess the learnability, usability, and workload characteristics of the proposed AR system. Six participants were recruited through convenience sampling: five graduate-level Computer Science students from Florida Gulf Coast University and one Herc Rentals employee with direct familiarity with the CCR domain. This mixed participant group enabled both technically informed usability feedback and domain-relevant interpretation of the sustainability content.

Prior to participation, all individuals provided written informed consent, were informed of their right to withdraw at any stage, and were advised that their responses would be anonymized and reported only in aggregate form. The study was conducted in accordance with the ethical guidelines of Florida Gulf Coast University.

The prototype was deployed as a release APK on a *Google Pixel 9 Pro XL* running *Android 16*. The application was developed in *Unity 6.2 (6000.2.3f1)* using *AR Foundation 6.2.0*. Participants first received a brief orientation to the application and then completed a sequence of five tasks: (1) launch the application and detect the printed Herc Rentals logo, (2) activate the three emission spheres, (3) inspect the information panel associated with each scope, (4) compare relative emission magnitudes and read the displayed equivalencies, and (5) reset the scene by toggling the logo again.

Following task completion, participants completed the System Usability Scale (SUS) and the NASA Task Load Index (NASA-TLX). SUS responses were collected using the standard five-point Likert format and converted to composite scores using the conventional scoring protocol, in which odd-numbered items contribute $(x - 1)$, even-numbered items contribute $(5 - x)$, and the summed score is multiplied by 2.5. NASA-TLX responses were recorded on the standard 21-point scale over six dimensions: Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration. To aid participant interpretation, the endpoints were labeled *Very Low* (0) and *Very High* (20), with intermediate integer values left available for fine-grained reporting.

The resulting SUS and NASA-TLX scores were used to assess whether the proposed interface could be operated effectively without prior training and whether the AR interaction imposed excessive cognitive or physical workload during use.

IV. Results and Discussion

A. System Usability Scale Results

The individual and aggregate System Usability Scale (SUS) results are summarized in Table III. The prototype achieved a mean SUS score of 77.9 with a standard deviation of 10.1. According to the interpretive guidelines reported by Bangor *et al.*, this score falls within the *Good* usability range and

exceeds the project target score of 75. Because the sample size was small ($N = 6$), the reported values are descriptive and should not be interpreted inferentially.

At the participant level, four of the six participants achieved scores at or above the target threshold of 75. Two participants ($P2 = 92.5$, $P3 = 90.0$) scored in the *Excellent* range, while two additional participants ($P4 = 75.0$, $P5 = 75.0$) met the target and remained within the *Good* range. Two participants scored below the target threshold: $P1 = 70.0$ and $P6 = 65.0$. These values still indicate acceptable usability, with $P1$ lying at the lower boundary of the *Good* band and $P6$ falling in the *OK* range. The comparatively lower score from the Herc Rentals employee ($P6$) may reflect stricter expectations from a domain-aware evaluator rather than a fundamentally negative interaction experience.

TABLE III: Individual and Mean SUS Scores ($N = 6$)

Participant	SUS	Rating	≥ 75
P1	70.0	OK	No
P2	92.5	Excellent	Yes
P3	90.0	Excellent	Yes
P4	75.0	Good	Yes
P5	75.0	Good	Yes
P6	65.0	OK	No
Mean	77.9	Good	4/6

Informal participant feedback further supported the quantitative findings. Several participants reported that the three-sphere metaphor made it immediately clear which emissions scope dominated the total footprint. Others noted that the equivalency views, particularly tree and diesel comparisons, made otherwise abstract metric-ton values more tangible and relatable. This observation is consistent with prior work showing that immersive unit-based visualization can improve comprehension of abstract magnitudes [4].

B. NASA-TLX Workload Results

The NASA Task Load Index (NASA-TLX) results are presented in Table IV. Overall, participants reported very low levels of Mental Demand, Physical Demand, Temporal Demand, and Effort, suggesting that the AR interaction imposed minimal cognitive or physical burden. These findings are consistent with the design goal of supporting intuitive interaction without requiring prior training.

Perceived Performance was rated at the maximum level by five of the six participants, indicating a strong sense of successful task completion. One participant ($P6$) provided a lower Performance rating of 12 on the 0–20 scale, which still corresponds to a relatively high perceived level of task success. Frustration scores were also uniformly low, with five participants reporting the minimum level and one participant ($P2$) reporting a value of 5, indicating only minor navigational difficulty. Taken together, these findings suggest that the system was easy to use, imposed limited workload, and did not create notable frustration during interaction.

The Herc Rentals employee ($P6$) produced the lowest SUS score and was also the only participant to provide a sub-maximum Performance rating. One plausible interpretation is

TABLE IV: NASA-TLX Results by Subscale ($N = 6$)

Participant	Mental	Physical	Temporal	Performance	Effort	Frustration
P1	V. Low	V. Low	V. Low	V. High	V. Low	V. Low
P2	V. Low	V. Low	V. Low	V. High	V. Low	5
P3	V. Low	V. Low	V. Low	V. High	V. Low	V. Low
P4	V. Low	V. Low	V. Low	V. High	V. Low	V. Low
P5	V. Low	V. Low	V. Low	V. High	V. Low	V. Low
P6	V. Low	V. Low	V. Low	12	V. Low	V. Low
Overall	V. Low	V. Low	V. Low	V. High*	V. Low	V. Low†

that domain-aware users may apply stricter standards regarding data completeness, interface precision, or organizational relevance when evaluating such systems. Importantly, however, $P6$'s Frustration rating remained very low, suggesting that the lower usability score is better understood as reflecting elevated expectations rather than a negative or unsuccessful interaction experience. Fig. 2 illustrates the AR prototype in operation, where three emission spheres are anchored to the detected Herc Rentals logo and presented in the user's physical environment.



Fig. 2: AR prototype in operation: three scope emission spheres anchored to the Herc Rentals logo.

C. Discussion

The findings provide preliminary support for the research questions posed in this work. First, *RQ1* is provisionally supported: participants were able to identify Scope 3 as the dominant emissions source directly from the bubble visualization, without relying on tabular data. Although this observation suggests that the AR representation improved immediate interpretability of relative emission magnitude, the absence of a controlled baseline condition prevents a causal attribution of this advantage to the AR modality alone. Second, *RQ2* is supported by the usability results. The prototype achieved a mean SUS score of 77.9, exceeding both the project target of 75 and the commonly accepted *Good* usability threshold. In addition, four of the six participants scored at or above the target threshold, and all participants completed the assigned interaction tasks without assistance. Third, *RQ3* is also supported: participant feedback indicated that the offset equivalencies made large emissions values more tangible and interpretable than raw metric-ton figures alone. The uniformly low NASA-TLX ratings further suggest that the tap-to-reveal interaction design imposed minimal cognitive and physical

workload, making the system appropriate for non-technical stakeholder audiences.

From a visualization perspective, the bubble-scaling metaphor effectively conveyed relative scope magnitude, with the contrast between Scope 3 (183,425 MT CO_2e) and Scope 2 (13,781 MT CO_2e) making value-chain dominance immediately apparent. Offset equivalencies, such as tree-based comparisons, also provided a more concrete framing of abstract carbon quantities, aligning with prior work showing that embodied or contextualized representations can enhance engagement [1].

Several limitations qualify these findings. The small sample size ($N = 6$) limits generalizability, and the absence of a comparison condition prevents causal claims about AR-specific benefits. System performance was not benchmarked, and future work should measure tracking stability, latency, and robustness across devices. The prototype also relied on locally stored JSON data, leaving issues of live data authenticity and enterprise integration unaddressed. Finally, information panels remain text-heavy and limited to a single organization; future iterations could incorporate multi-year transitions, broader datasets, and live sustainability-platform connectivity. These results provide an encouraging basis for future controlled studies comparing AR-based sustainability reporting with conventional formats to determine whether observed interpretability and engagement benefits stem from AR, visual encoding choices, or their combination.

V. Conclusion

This paper presented an interactive augmented reality system for visualizing corporate carbon footprints through dynamic Scope 1, Scope 2, and Scope 3 emission spheres anchored to a physical organizational logo. Using Herc Rentals Corporate Citizenship Report data from 2022–2024 together with U.S. EPA greenhouse gas equivalency factors, the system converts static sustainability disclosure into a spatially grounded and interactive experience. A formative evaluation with six participants yielded a mean SUS score of 77.9, exceeding the target threshold of 75, while NASA-TLX responses indicated generally low cognitive and physical workload. Although preliminary, these results suggest that AR-based sustainability communication can be learnable, accessible, and helpful for non-expert interpretation of corporate emissions data. Future work will extend the system with animated multi-year views, richer interaction modalities, enterprise data integration, and larger controlled studies. More broadly, this work highlights the potential of AR to make sustainability reporting more interpretable and engaging across corporate and public-sector contexts.

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